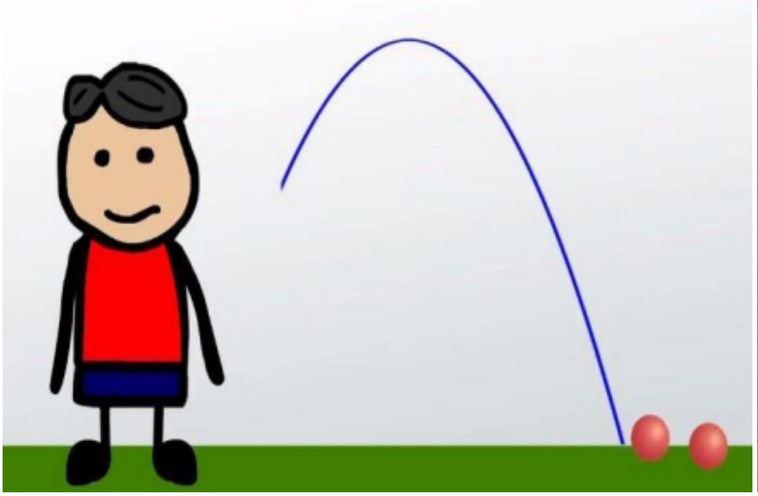
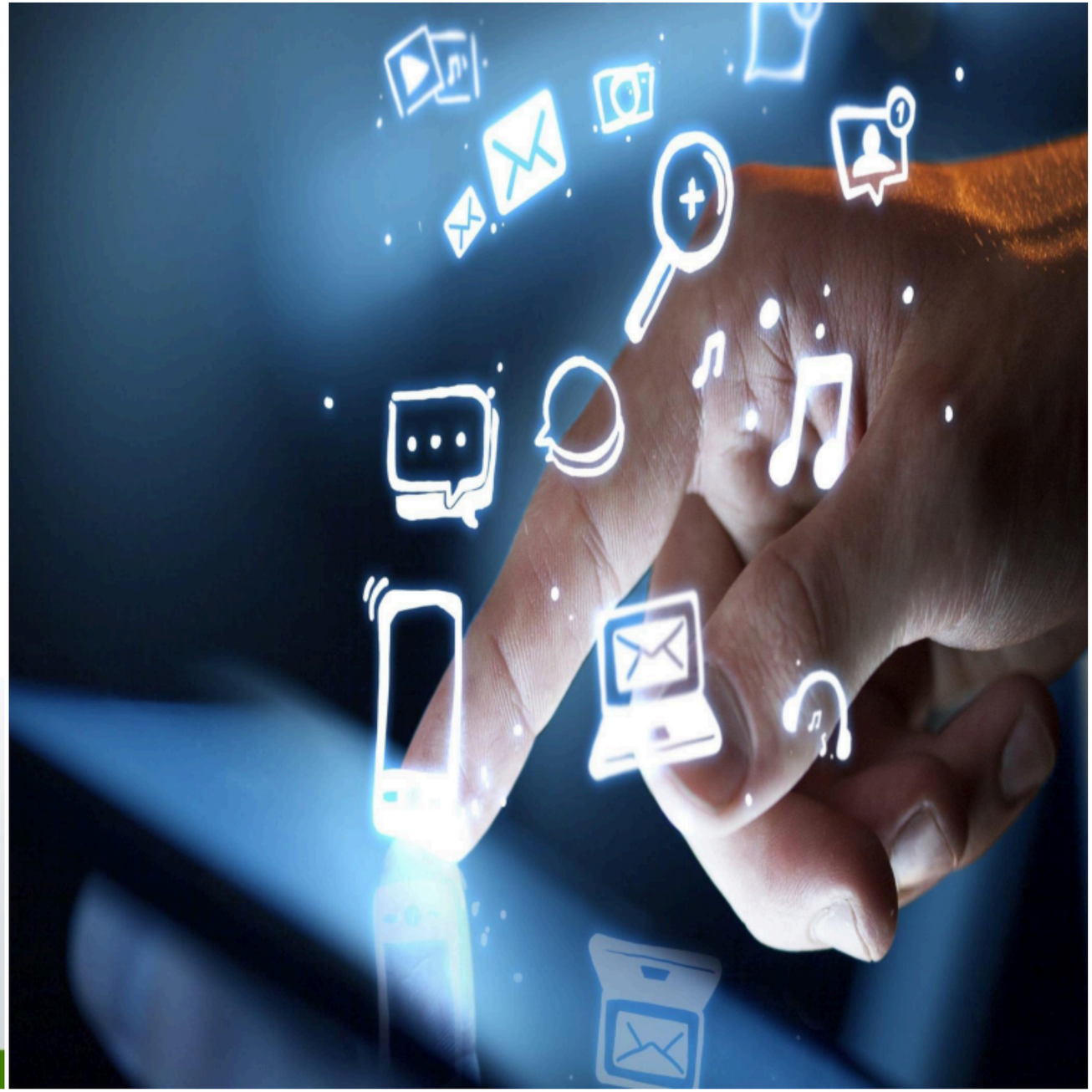




ROOTSOFEQUATION

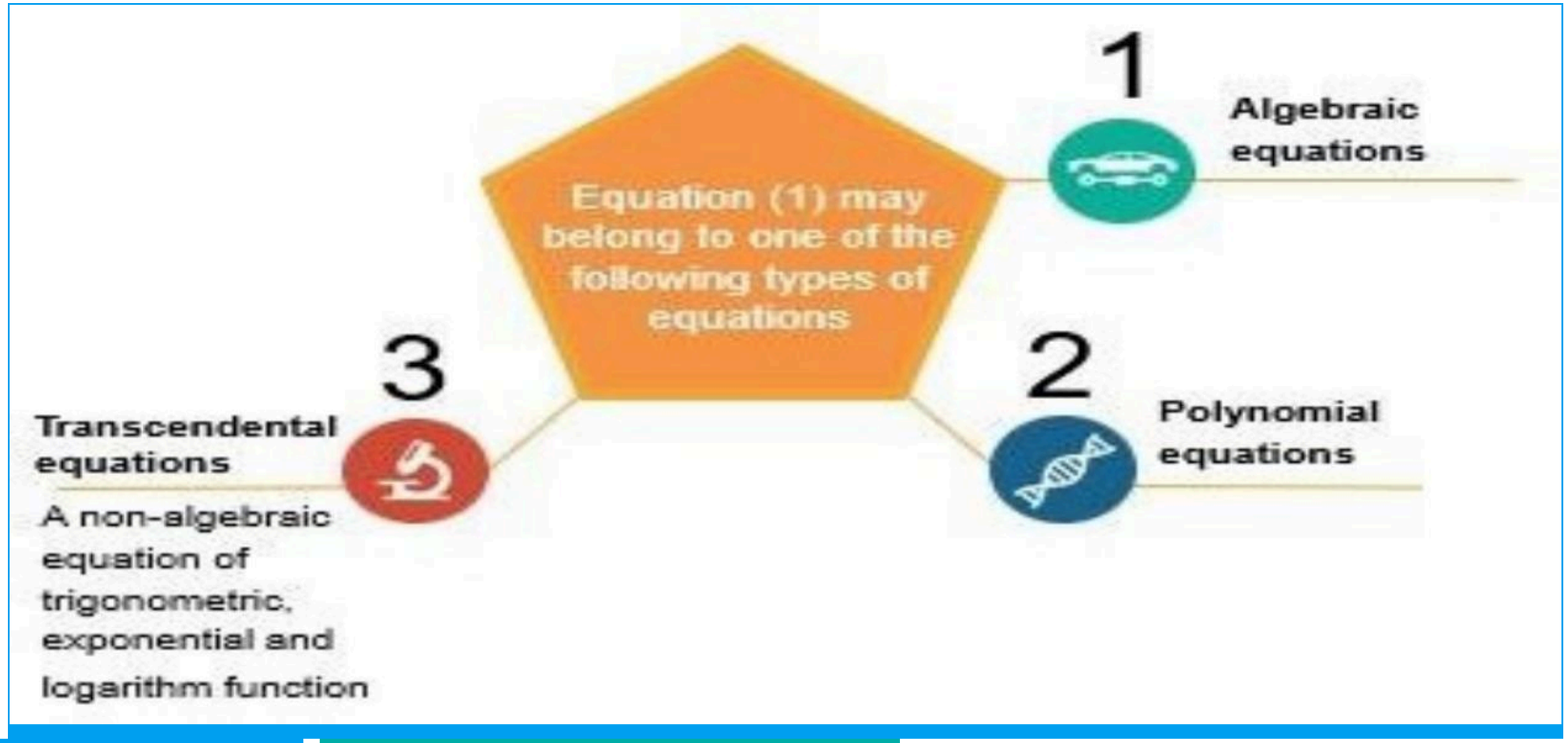
Introduction

❑Mathematical model in science and engineering involve equationsthat need to be solved.

❑ Equation of one variable can be formulated as 𝑓(𝑥) = 0 (1)

❑ Equation (1) can be in the form of linear and nonlinear. ❑ Solving equation (1) means that finding the values of *x* that satisfyingequation (1).

Introduction



Introduction

Example 1: Algebraic Equation

$$4x - 3x^2 - 15 = 0$$

Example 2: Polynomial Equation

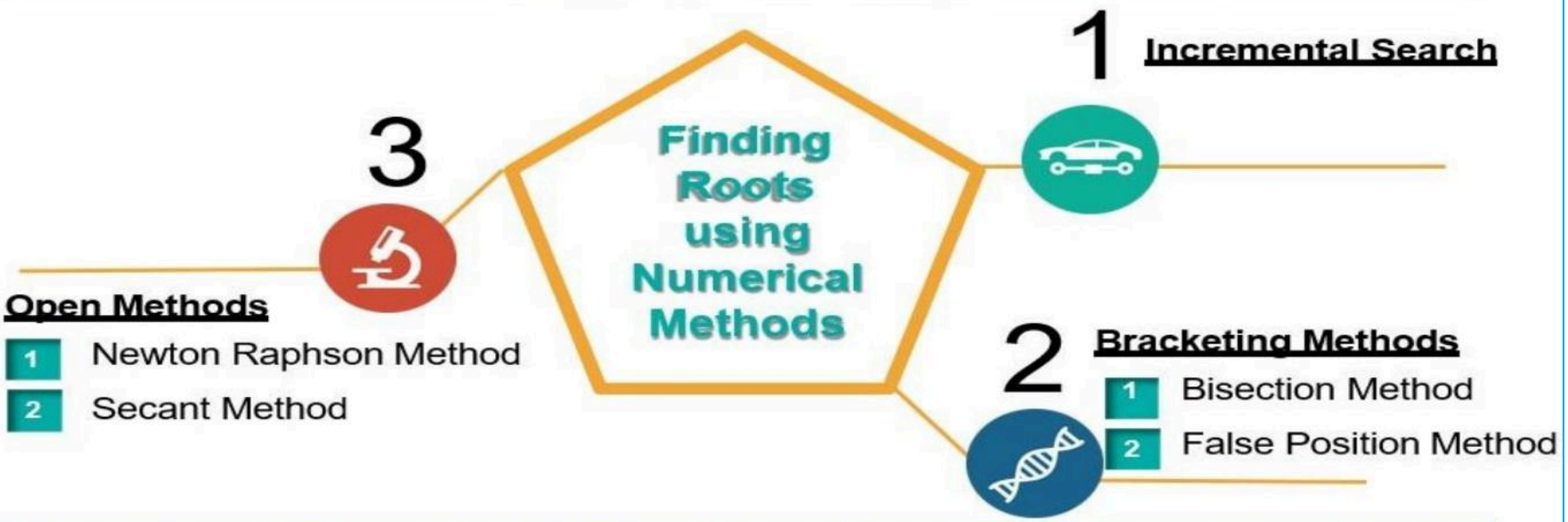
$$x^2 + 2x - 4 = 0$$

Example 3: Transcendental Equation

$$\sin(2x) - 3x = 0$$

Introduction

Three types of Numerical Methods shall be considered to find the roots of the equations:



Prior to the numerical methods, a **graphical method** of finding roots of the equations are presented.

Graphical Method

❑ Graphical method is the simplest method ❑ The given function is plotted on Cartesian coordinateand

x−values (roots) that satisfying $f(x) = 0$ is identified. ❑ x −values (roots) satisfying $f(x) = 0$ provide approximationrootsforthe underlying equations.

❑ $f(x)$ can have one or possibly many root(s).

Graphical Method

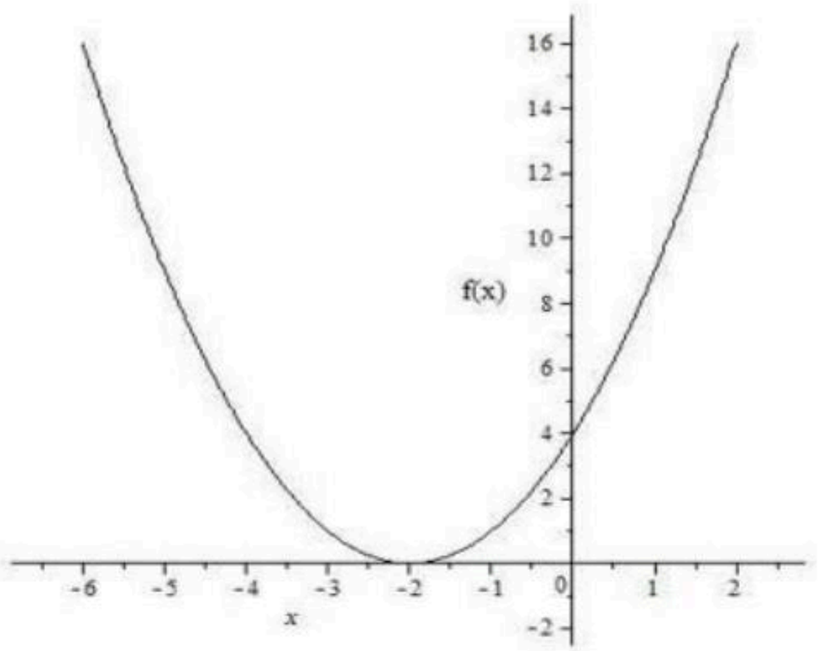


Figure 1 : One Solution

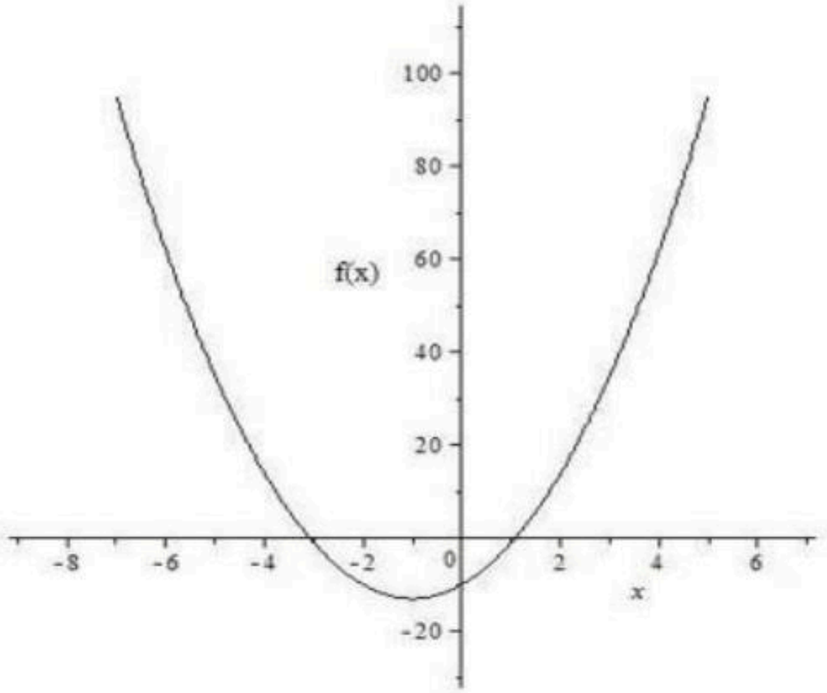


Figure 2 : Two Solutions

1. Incremental Search

❑ Incremental search is a technique of calculating ◆ (x) for incremental values

of *x*- over the interval where the root lies. ❑ It starts with an initial value, x_0.

❑ The next value x_i for i = 1,2,3, … is calculated by using $x_i = x_{i-1} + h$

where h is referred to a step size.

❑ If the sign of two x (x) changes or if

$$f(x_n) \cdot f(x_{n-1}) < 0$$

then the root exist over the prescribed interval of the lower bound, x_l and upper bound, x_u.

❑ The root is estimated by using

$$x_r = \frac{x_l + x_u}{2}$$

Incremental Search

Example

Find the first root of $f(x) = 4.15x^2 - 16x + 8$ by using incremental search. Start the procedure with the initial value, $x_0 = 0$ and the step size, $h = 0.1$. Perform three iterations of the incremental search to achieve the best approximation root.

Solution

Start the estimation with initial value $x_0 = 0$ and step size, $h = 0.1$.

x	$f(x)$
0	8
0.1	6.4415
0.2	4.966
0.3	3.5735
0.4	2.264
0.5	1.0375
0.6	-0.106



$$f(0.5) \cdot f(0.6) < 0$$

$$x_r = \frac{0.5 + 0.6}{2} = 0.55$$

Incremental Search

Solution (Cont.)

Increasing the accuracy of root estimation with step size, $h = 0.01$ for $x \in [0.5, 0.6]$

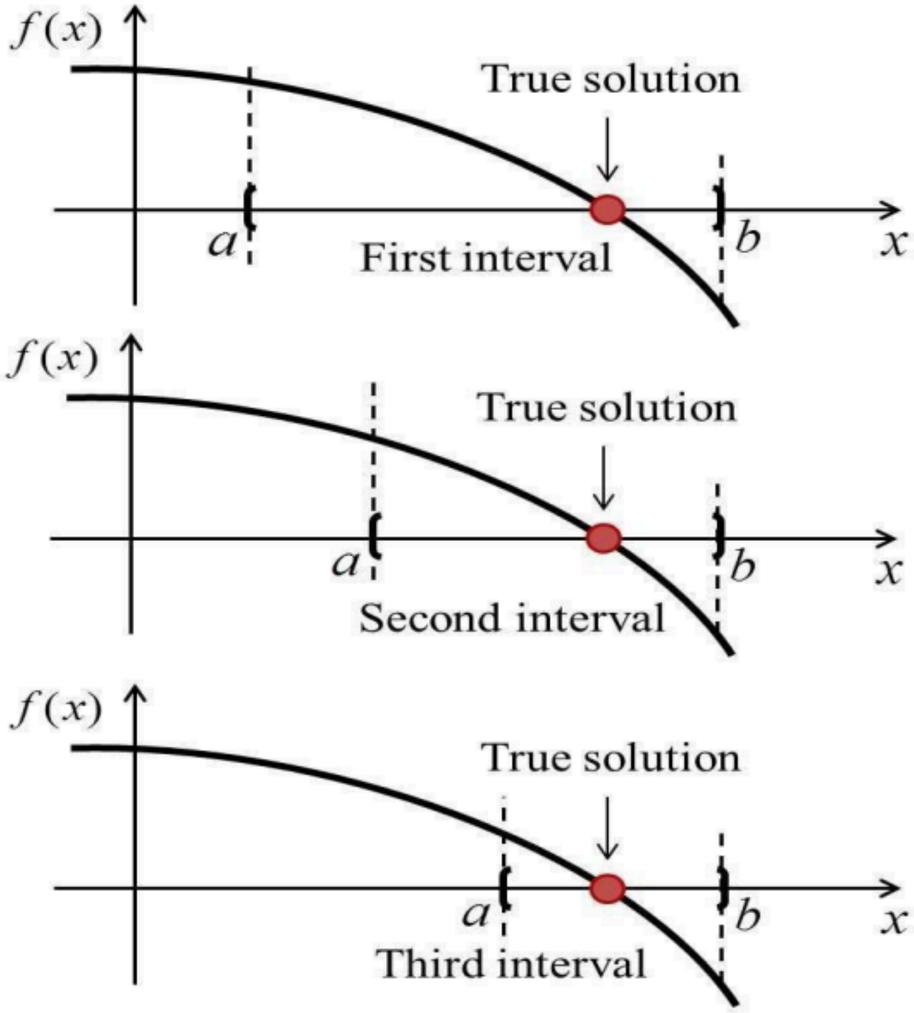
x	$f(x)$
0.5	1.0375
0.51	0.919415
0.52	0.80216
0.53	0.685735
0.54	0.57014
0.56	0.455375
0.57	0.34144
0.58	0.11606
0.59	0.004615
0.60	-0.106

$$f(0.59) \cdot f(0.6) < 0$$

$$x_r = \frac{0.59 + 0.6}{2} = 0.5950$$

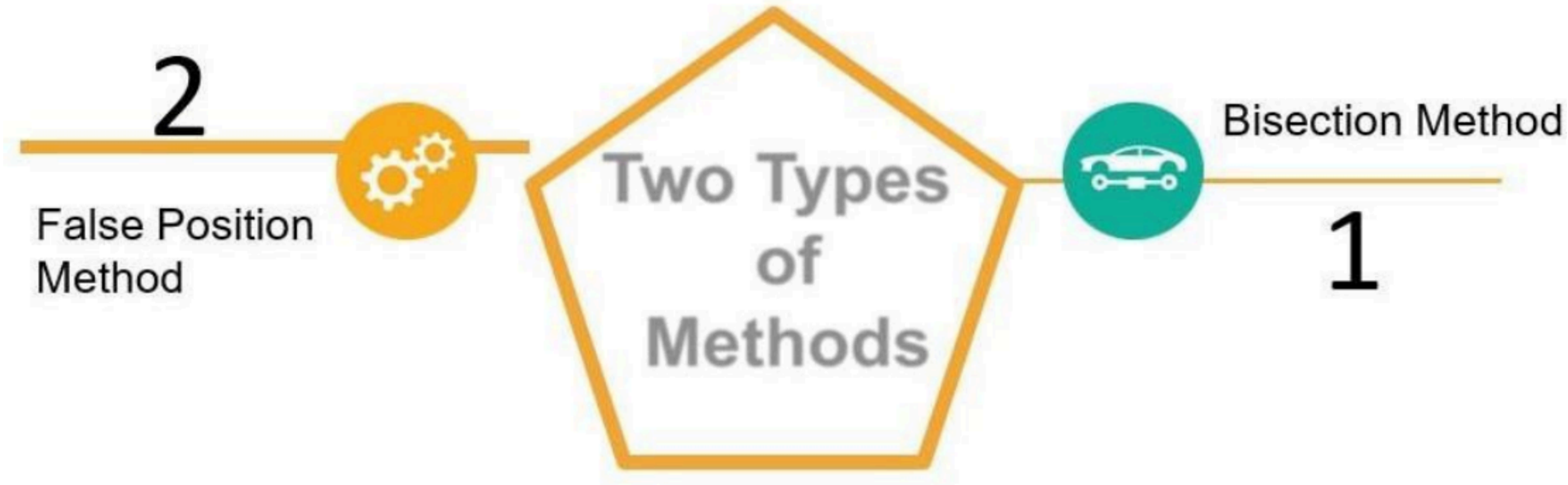


- The figure illustrates the basic idea of bracketing method–that is guessing an interval containing the root(s) of a function.
- Starting point of the interval is a lower bound, x_l .End point of the interval is an upper bound, x_u .
- By using bracketing methods, the interval will split into two subintervals and the size of the interval is successively reduced to a smaller interval.
- The subintervals will reduce the range of intervals until its distance is less than the desired accuracy of the solution



Two Types of Bracketing Methods

- Bisection Method
- False Position Method



Bisection Method

- This is a very simple method. Identify two points x = a andx=bsuchthatf(a)and f (b) are having opposite signs. Let *f* (a) be negativeand*f* (b)bepositive. Then there

will be a root of $f(x) = 0$ in between aandb.

- Let the first approximation be the mid point of the interval (a, b). i.e.

$$x_1 = \frac{(a+b)}{2}$$

- If $f(x_1) = 0$, then x_1 is a root, otherwise root lies between aand$x_1$or$x_1$andb according as $f(x_1)$ is positive or negative. Then againwebisecttheinterval and continue the process until the root is found todesiredaccuracy.

Let $f(x_1)$ is positive, then root lies in between a and x_1 (seefig. 1).Thesecond approximation to the root is given by,

$$x_2 = \frac{(a+x_1)}{2}$$

Bisection Method

- If $f(x_2)$ is negative, then next approximation is given by

$$x_3 = \frac{(x_2 + x_1)}{2}$$

Bisection Method

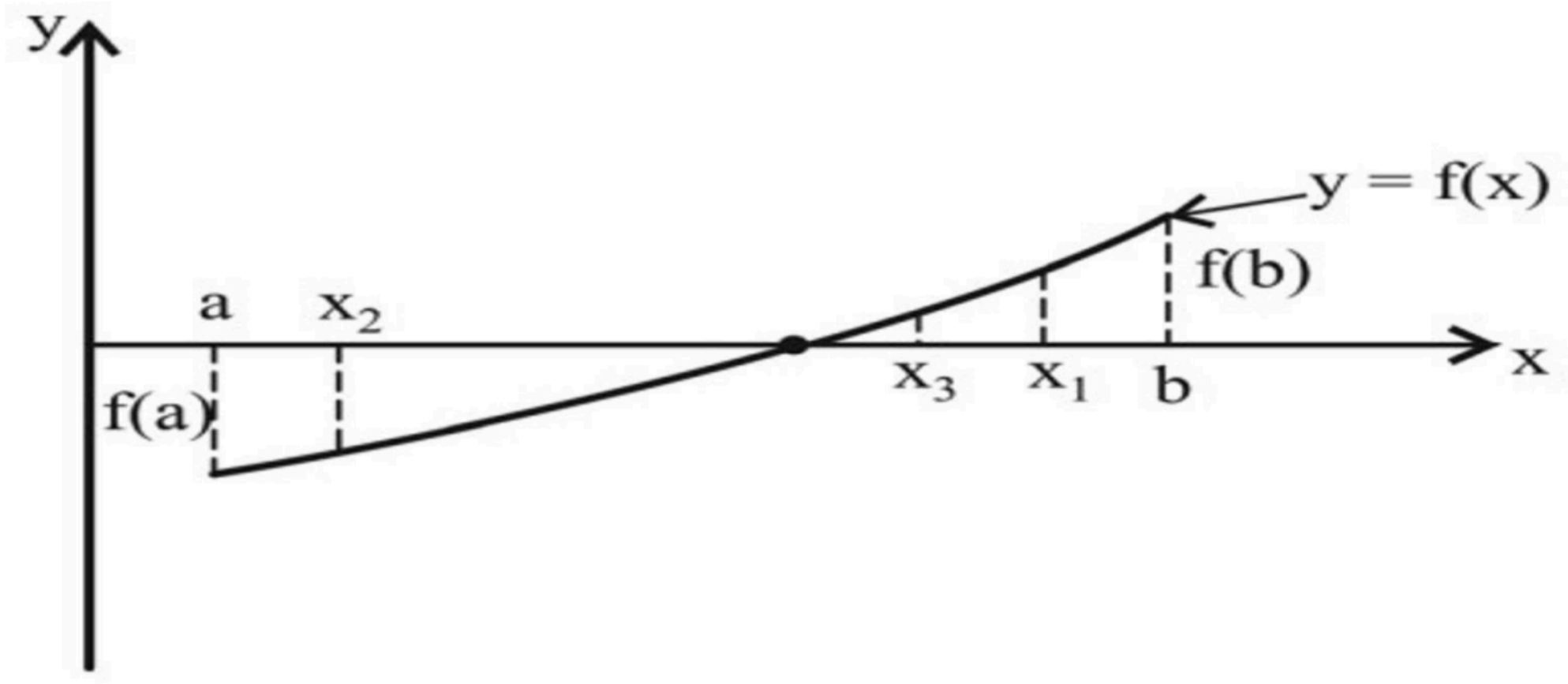


Figure 1. Bisection Method

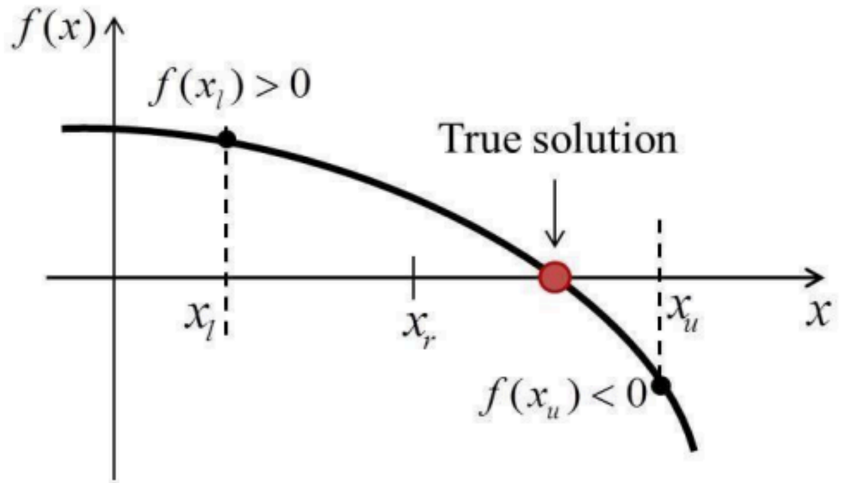
Bisection Method

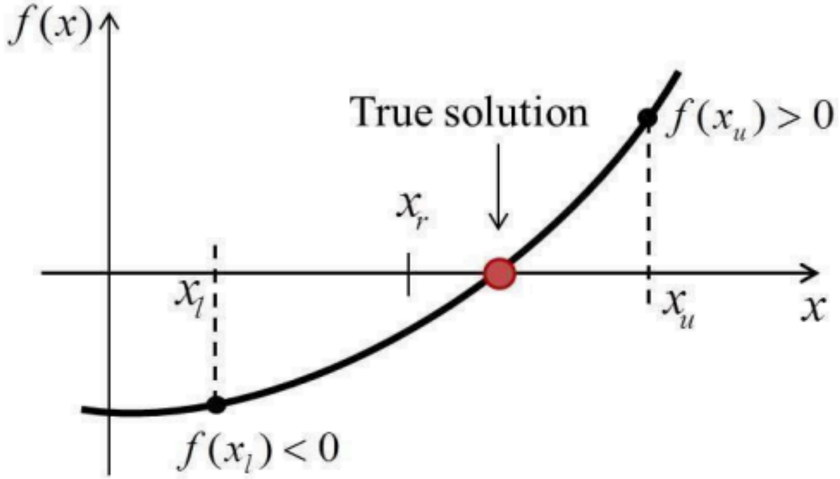
❑ Bisection method is the simplest bracketing method. ❑ The lower value, x_l and the upper value, x_u which bracket theroot(s)are required.

❑ The procedure starts by finding the interval [x_l , x_u] wherethesolutionexist.

❑ As shown in =

, at least one root exist in the interval [x_l ,x_u] if $f(x_l)$

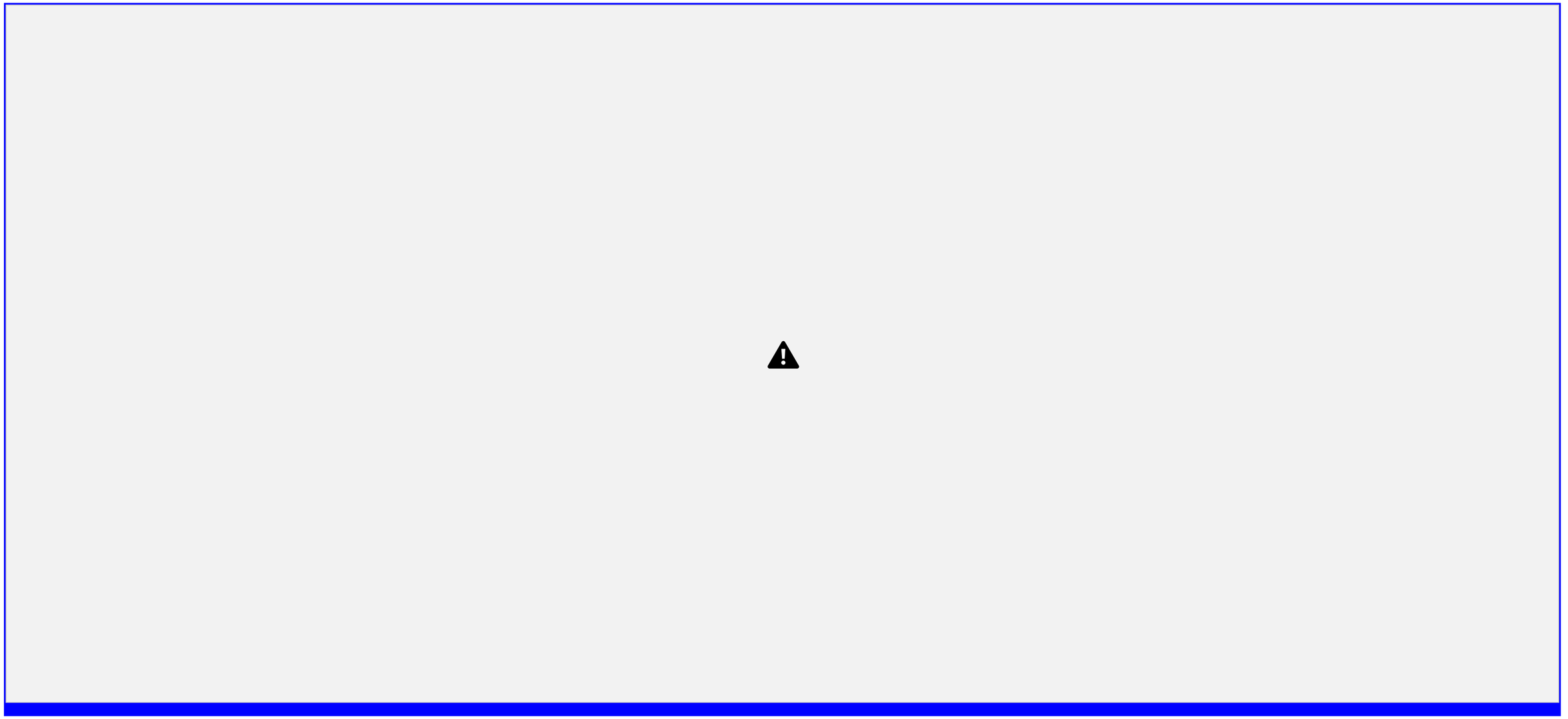




$\cdot f(x_u) < 0$ Solution of $f(x) = 0$

Bisection Method

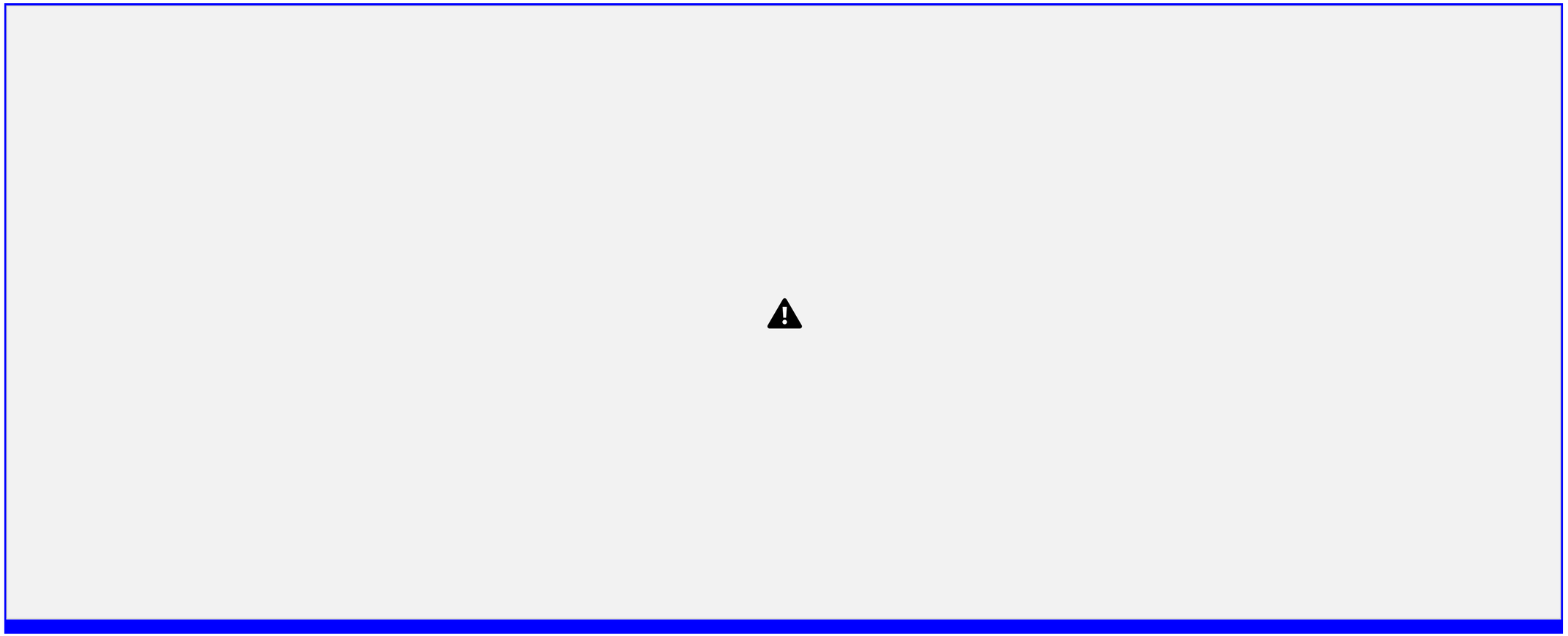
Algorithm



Example 1

Find a real root of the equation $f(x) = x^3 - x - 1 = 0$, using

Bisectionmethod.**Solution**



Example 1



Example 2

Find the root of the function $f(x) = x^3 + 3x - 5$ using bisection method. **Solution**

$f(0) = 0 + 0 - 5 = -5$
$f(1) = 1 + 3 - 5 = -1$
$f(2) = 8 + 6 - 5 = 9$

Example 2

Solution

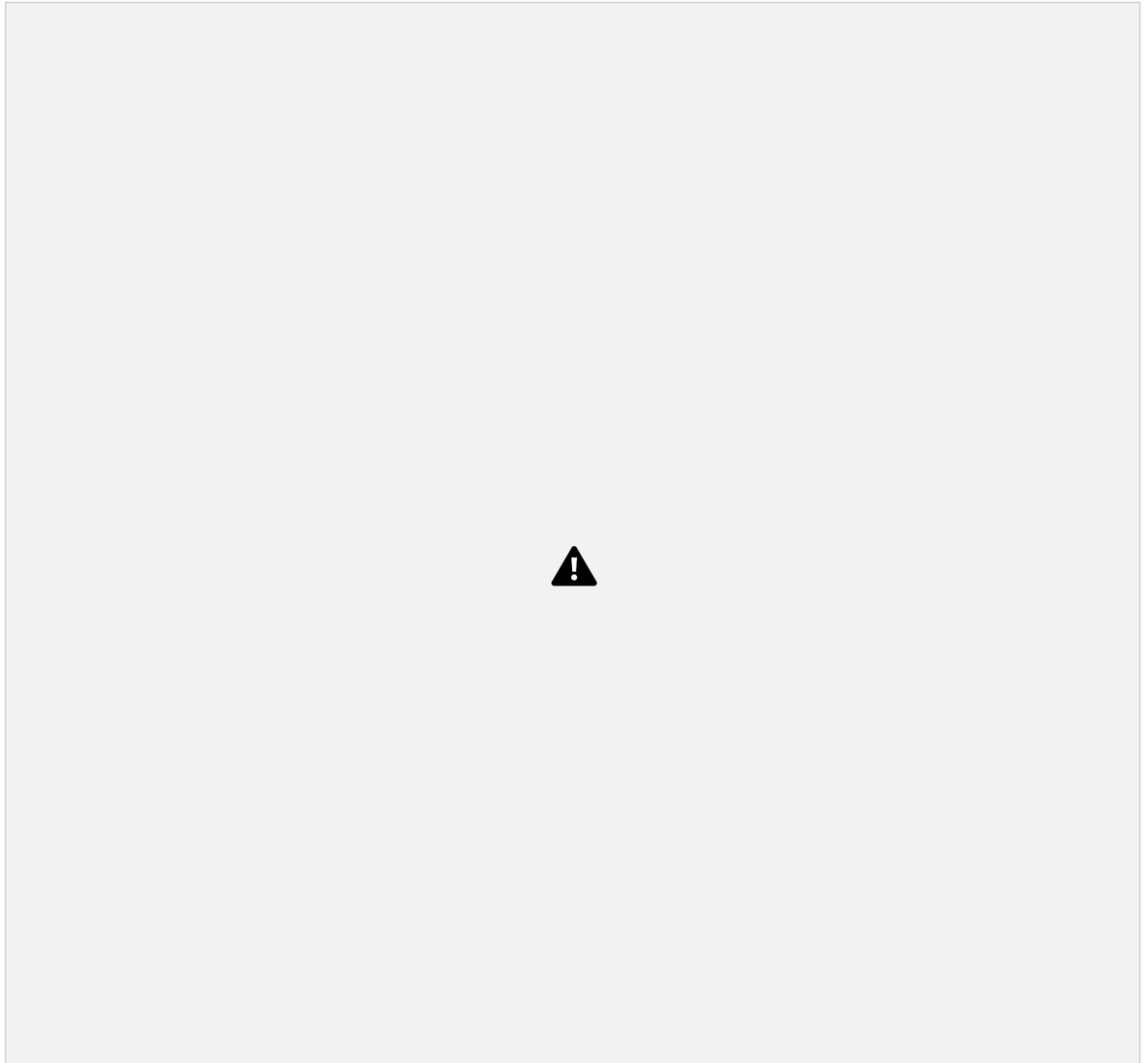
a		f(a)	f(b)		(a+b)/2f(c)	b-a
1					2	
1			-1		1.25 0.7	
1			-1		-0.20	
1.125		1.25 -		0.703 1		0.2370.125
1.125	1		-0.201	0.237 1.1		0.0140.0625
1.125	1.1		-0.201	0.014 1		-0.0940.03125
1.1406	1.1		-0.094	0.014 1		-0.0400.0156
1.1484	1.1		-0.040	0.014 1		0.0078

since (b-a) at b=1.15625 is less than 0.01 therefore the approximateroot of thegiven function is 1.15625.

False Position Method

- It is an improvement of the Bisection method.
- The bisection methodconverges slowly due to its behavior in redefined the size of interval that containing the root.
- The procedure begins by finding an initial interval x_l, x_u that bracket the root.
- $f(x_l)$ and $f(x_u)$ are then connected using a straight line.

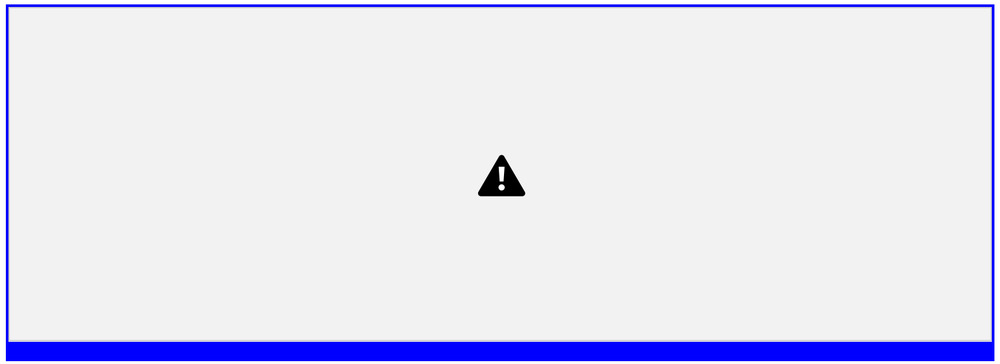
  The estimated root, x_r is the x



- value where the straight line crosses ? -axis.

False Position Method

False Position Method Formula



Example 1

Find the root of the equation f (x) = $3x^3 – x – 1 = 0$ by falsepositionmethod using at least 4 iterations.

Solution

$f(0) = 0 - 0 – 1 = -1$

$f(1) = 3(1)^3 - 1 – 1 = 1$

Example 1

Solution

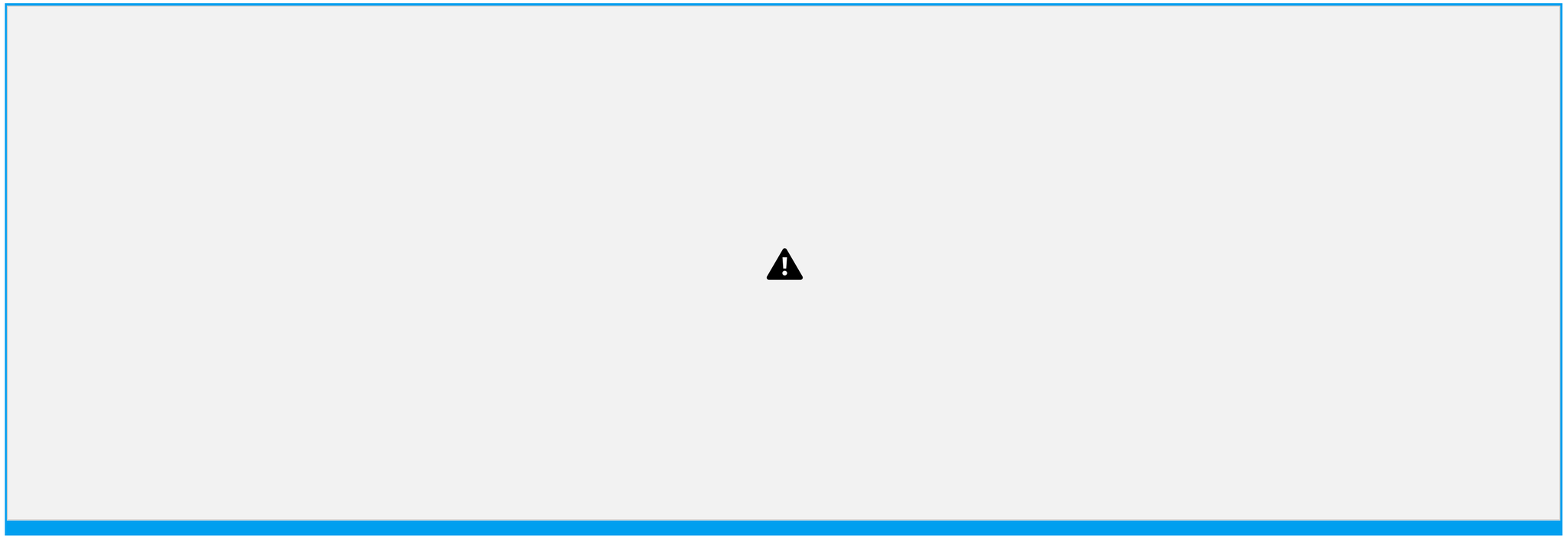
x_L	x_u	$f(x_L)$	$f(x_u)$	⚠	$f(x_m)$
0					-1.12
0.5		-1.		1 (	-0.4
0.765		-0.		1 (	-0.0

0.835		-0.		1	-0.01

3. Open Methods

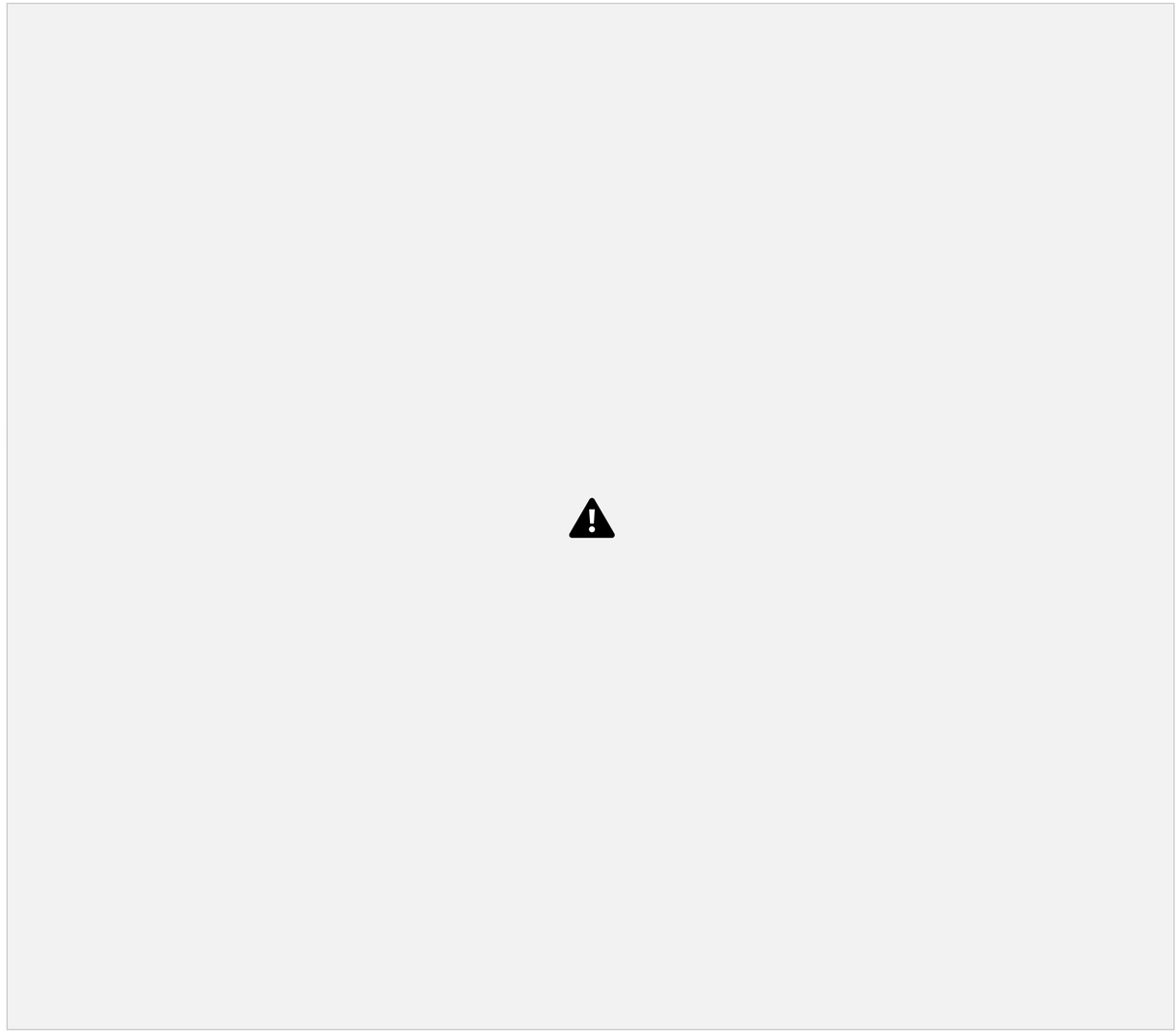
- The idea of this method is to consider at least one initial guesswhichisnot necessarily bracket the root.
- Normally, the chosen initial value(s) must be close to theactual rootthatcan be found by plotting the given function against itsindependentvariable.
- In every step of root improvement, ⬦ ⬦ of previous step is consideredasthe previous value for the present step.
- In general, open methods provides no guarantee of convergencetothetrue value, but once it is converge, it will converge faster thanbracketingmethods.

Two Types of Open Methods



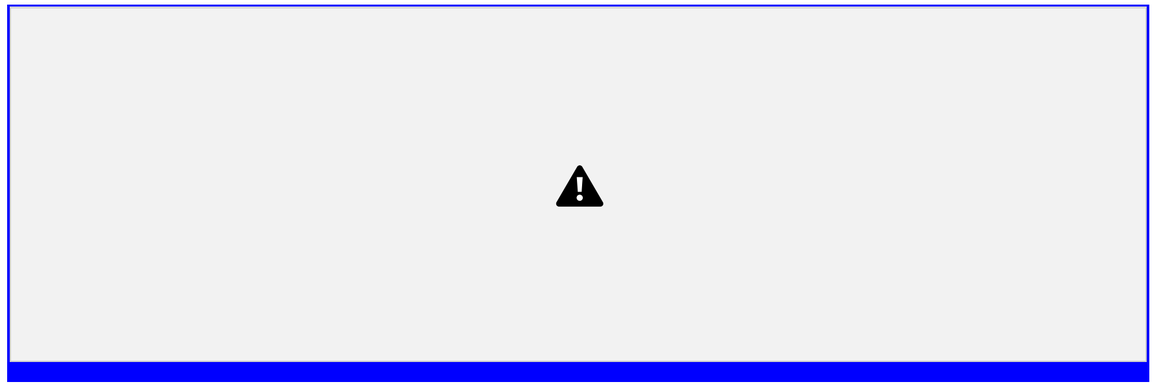
Newton Raphson Method

- It is an open method for finding roots of $f(x) = 0$ by using the successive slope of the tangent line.
- The Newton Raphson method is applicable if $f(x)$ is continuous and differentiable.
- Numerical scheme starts by choosing the initial point, x_0 as the first estimation of the solution.
- The improvement of the estimation of x_1 is obtained by taking the tangent line to $f(x)$ at the point x_0, $f(x_0)$ and extrapolate the tangent line to find the point of intersection with an x – axis.



Newton Raphson Method

Newton Raphson MethodFormula



Example 1

Determine the root of the equation $f(x) = 3x^3 – x –1= 0$byusingnewtonraphson method at least four iteration.

Solution

$f(x) = 3x^3 – x –$

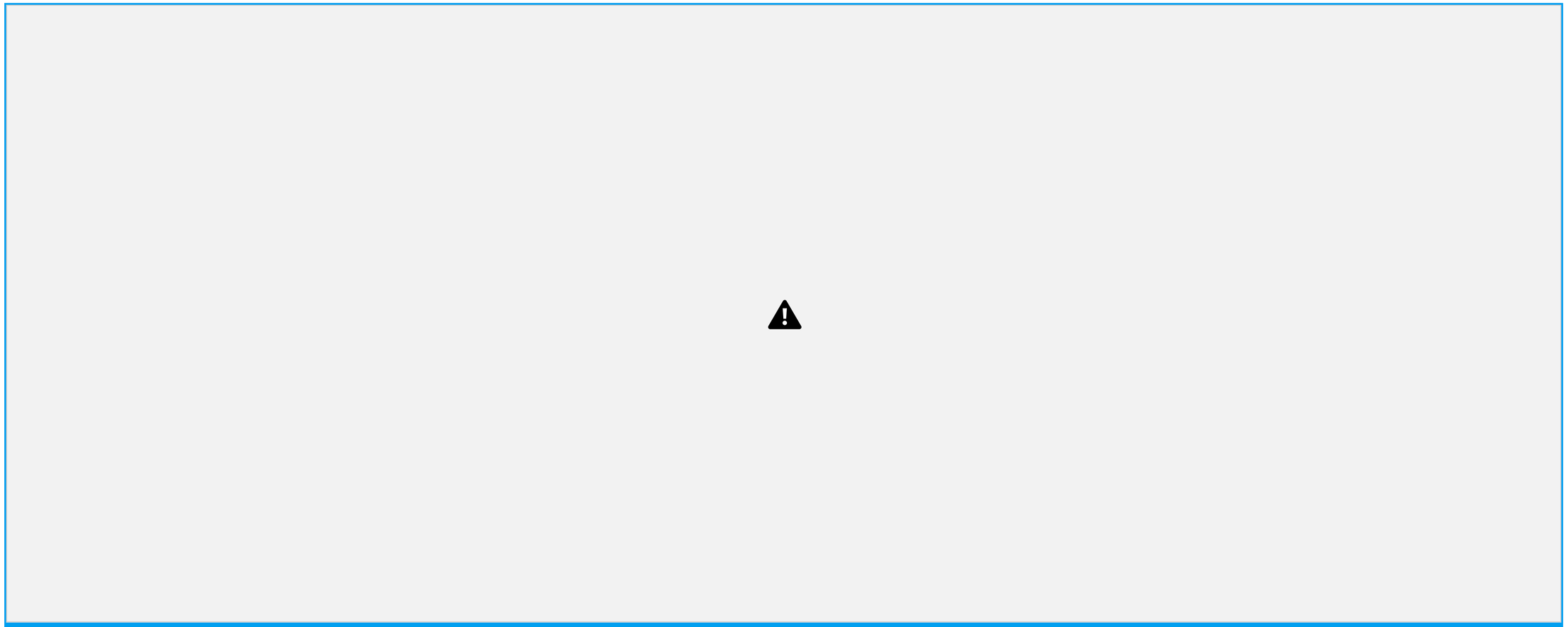
$1 = 0$

i	x_i	X_{i+1}
1		10.875
2		0.8750.8521
3		0.85210.8514
		0.85140.8513

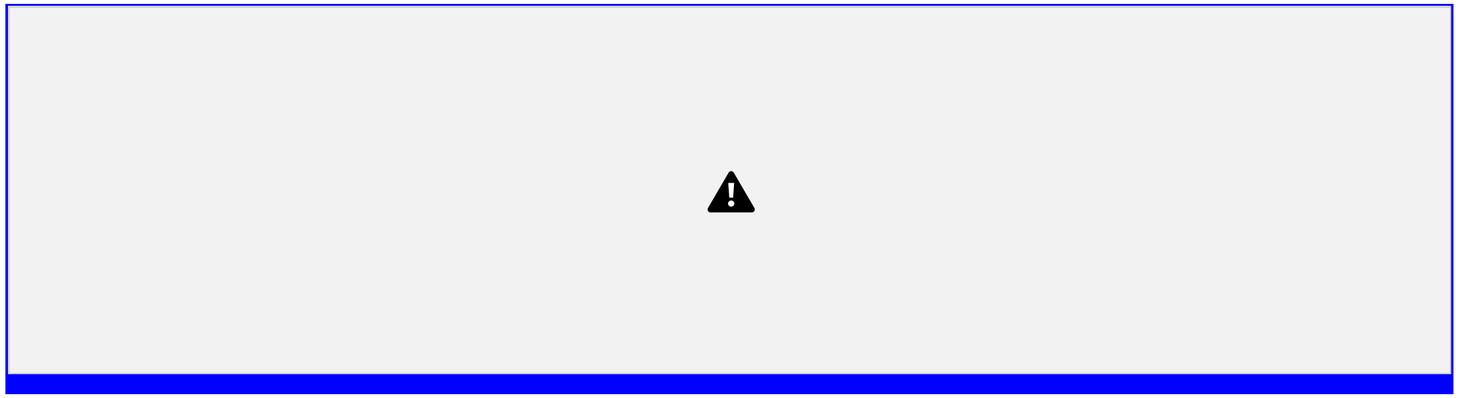
$f'(x) = 9x^2 - 1$

$\frac{3(x_i)_3 - i}{i} - 1 \quad 9(x_i)_2 - 1$

Secant Method



Secant Method Formula Secant Method Formula



= 0 by secant method using at least 4 iterations.

Solution

$f(x) = x^2 - x -$

i	x
1	
2	

Example 1

Find the root of the equation $f(x) = x^2 - x - 1$

3	
4	

$x_a = 3$

$x_b = 2$



1 = 0

Conclusion

Need two initial guesses Can involve one or more initial guessesThe root is located within an interval

Not necessarily bracket theroot.

prescribed by a lower and an upper

bound.

Always work but converge slowly Do not always work (can diverge) but when they do they usually converge much more quickly.

ThankYou!!!